Circular Economy I: Business Models & Regenerative Design

Week 4: Phase II - Circular Architecture & Operations

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FINANCIAL ECONOMICS

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Roadmap & Learning Objectives

Today's Flight Plan

1. The Problem
Take-Make-Waste Liability

2. Methodology
Butterfly Diagram & 9Rs

3. The Solution
C2C & Regenerative Models

4. Simulation
Redesigning for Profit

Executive Objectives

- Differentiate between *technical* and *biological* cycles to allocate capital efficiently.
- Apply the **9Rs framework** to re-engineer product lines for maximum resource productivity.
- Transform traditional waste management costs into **new revenue streams (ROI)**.

The Disassembly Audit

Let's do some math. *Imagine the financial lifecycle of a standard $999 smartphone.*

Capital Allocation

$80

Raw Materials Cost

$150

Manufacturing & R&D

The Linear Trap

⚠ Financial Hemorrhage

A linear economy forces companies to constantly buy virgin materials at volatile market prices, only to let consumers send them to a landfill.

We are literally throwing our working capital into the trash.

Question: When the battery dies in 2 years and the consumer throws it away, *who absorbs the loss of those rare earth metals?*

Waste as a Design Flaw

" Waste is a human invention. In nature, the waste of one system is the food for another. "

Braungart & McDonough (Cradle to Cradle)

$\rightarrow$ **The ADE Paradigm Shift:**
If a product requires a landfill or an incinerator at the end of its life, it is a defective product. Every gram of unrecoverable waste is a direct drag on your Cost of Goods Sold (COGS).

The Ellen MacArthur Butterfly Diagram

The Core Rule of Circularity: *Never mix the cycles.*

Biological Cycles

Materials that can safely re-enter the biosphere.

Action: Cascade, extract biochemicals, anaerobic digestion, compost.

Example: Organic cotton without toxic petroleum dyes.

Technical Cycles

Materials that cannot biodegrade and must remain in industrial loops.

Action: Share, Maintain, Reuse, Remanufacture, Recycle.

Example: Lithium, aluminum, specialized polymers.

The Enemy: Monstrous Hybrids.

A leather shoe (biological) glued with polyurethane (technical). It cannot be composted, nor can it be melted down. **Residual Value = Zero.**

Why a "Butterfly"? Anatomy of the Framework

The Metaphor: *Before looking at the full complex diagram, understand its architecture. The visual structure explains the exact financial efficiency of the loops.*

The "Body" (Linear)

The central spine represents the linear flow: Extract → Produce → Consume → Waste.
The goal is to starve the body.

The "Wings" (Circular)

Left wing = Biological Cycles.
Right wing = Technical Cycles.
These loops circulate materials back into the economy, preventing them from hitting the landfill.

The Golden Rule of Proximity

The tighter the loop (the closer it is to the center of the butterfly), the more embedded value is retained, and the **higher the ROI**.

Business Example:

Repairing a smartphone (a very tight loop) is vastly more profitable and energy-efficient than grinding it down to *recycle* the raw metals (the widest loop).

The 9Rs Framework: A Hierarchy of Value Retention

Not all "R"s are created equal. The lower you go, the more capital you waste.

Smarter Use

(Highest ROI & Margins)

R0 Refuse:
Make product redundant (Digitization).

R1 Rethink:
Product-as-a-Service (PaaS).

R2 Reduce:
Maximize manufacturing efficiency.

Extend Life

(Mid ROI - New Markets)

R3 Reuse:
Second-hand markets.

R4 Repair:
Selling spare parts/services.

R5 Refurbish:
Restore to updated standards.

R6 Remanufacture:
Use old parts in new products.

Useful App.

(High Cost / Value Loss)

R7 Repurpose:
Use for a different function.

R8 Recycle:
Process materials (loses quality).

R9 Recover:
Incineration for energy (The absolute bottom).

Deep Dive 1: Smarter Use (R0-R2)

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The PaaS Revolution (Product-as-a-Service)

Core Executive Idea: Completely disconnecting revenue growth from physical resource consumption. You sell the *result*, not the *object*.

The Example: Michelin

Action (R1 Rethink):
"Tires-as-a-Service". Fleet managers pay per kilometer driven, not per physical tire.

Business Value:
Michelin is now financially incentivized to make tires that *never* wear out. Total alignment between sustainability and profit margins.

The Risk: Jevons Paradox

Action:
Car-sharing apps (Uber/Cabify) claiming R1 (Rethink/Sharing).

The Reality:
While individual car ownership drops, the *total* miles driven and congestion often increase, leading to higher overall emissions.

Nuance: Efficiency does not equal Absolute Reduction.

Deep Dive 2: Extend Lifespan (R3-R6)

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The Cannibalization Myth

Does repairing or refurbishing our products destroy new sales? Or does it unlock a new demographic?

The Example: Back Market

Action (R5 Refurbish):
Creating a standardized, trusted B2C marketplace for refurbished electronics.

Business Value:
Captures a price-sensitive segment that would *never* buy a new $1,200 iPhone. It expands the Total Addressable Market (TAM) without cannibalizing premium flagship sales.

Nuance: The "R" Conflict

The Dilemma:
Can you have R3 (Reuse) and R2 (Reduce) simultaneously?

The Trade-off:
Designing a glass bottle to be reused 50 times (R3) makes it thicker and heavier, requiring more fuel to transport (failing R2).

You must calculate the exact *break-even point* via Life Cycle Assessment (LCA).

Case 1: Interface Carpets (The B2B Pioneer)

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The Shift from Product to Service (R1 Rethink)

Before (Linear)

✗ Sold massive carpet rolls.
✗ If coffee spilled, the client replaced the *whole* room.
✗ High dependency on virgin petroleum (nylon).

→

After (Circular)

✓ Invented carpet tiles (Modularity).
✓ Only replace the stained tile.
✓ Take back old tiles, recycle the nylon indefinitely (Closed Loop).

→ **The ROI Result: Margins increased. They locked B2B clients into long-term "floor-covering" contracts and mitigated supply chain volatility.**

Case 2: Patagonia & Fairphone (The B2C Models)

Patagonia (Worn Wear)

Strategy: R4 (Repair) and R3 (Reuse).

Execution: They buy back old gear, clean it, repair it, and resell it on their own platform.

Business Impact:
Created a secondary market where they control the margins. Extremely high brand loyalty (Lowest CAC).

Fairphone

Strategy: Design for Disassembly (DfD).

Execution: 100% Modular phones. Users can upgrade the camera or swap the battery with a standard screwdriver.

Business Impact:
Defies the Apple obsolescence model. Sells component upgrades directly to consumers (New recurring revenue line).

Boardroom Pitch

The P&L Restructuring Challenge

Executive Simulation: The Context

The Crisis (Spanish Market):

You are the Board of Directors of a legacy White-Goods brand (e.g., Balay/BSH) based in Zaragoza. The new Spanish *Ley de Residuos* (Tax on virgin plastics) and the EU Right-to-Repair directive are projecting a **35% drop in your profit margins** by 2027.

The Dynamics (25 mins):

The Setup

We will divide into **3 Specialized Task Forces** (Groups A, B, C). Each group will tackle the margin crisis using a different segment of the 9Rs framework.

The Deliverable

The Common Goal: Pitch a 5-minute strategy to the CEO (Me) proving how your specific approach will offset the incoming regulatory costs and generate ROI.

Task Force Assignments

Group A: Smarter Use

Focus: R0, R1, R2

Shift the business model. How do we sell "Washing" instead of "Washing Machines"? How do we retain ownership of the metal and plastic assets?

Group B: Extend Life

Focus: R3, R4, R5, R6

Re-engineer the after-sales. How do we monetize the repair network? How do we design the machine so parts can be easily swapped and upgraded by the user?

Group C: Recovery

Focus: R7, R8, R9

The reverse logistics. When the machine finally dies, how do we harvest the components (Material Banks) to drastically reduce our COGS for the next generation of products?

Debriefing: The Common Reflection

" Based on the three pitches, which strategy provides the fastest short-term liquidity, and which provides the strongest long-term moat against competitors? Can a company survive by only implementing Group [R7 to R9]] strategy? "

The Executive Synthesis

The Pareto 80/20: Executive Takeaways

1. The 80% Rule

Up to **80% of environmental impact** and end-of-life financial liability is locked in at the *design phase*.

Takeaway: You cannot market, sell, or recycle your way out of a fundamentally bad linear design.

2. P&L, Not CSR

In Spain, the *PERTE* and virgin plastic taxes (0.45€/kg) mean linear models are mathematically doomed.

Takeaway: Circularity is not philanthropy; it directly lowers your COGS and slashes Customer Acquisition Cost (CAC).

3. Beware Downcycling

If your brand's sustainability claim relies entirely on **R8 (Recycling)**, your margins are highly vulnerable to supply shocks.

Takeaway: The true value for ADE professionals lies in **R1 (Servitization)** and **R4 (Repair)**.

Thank You.

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